

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Summer 2026 (Supply.)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology
Semester : SEMESTER - 3

Subject Code & Name: 25AF1293PC302 - ELECTRICAL AND ELECTRONICS MEASUREMENTS

Time : 3(Hours)

[Total Marks : 60]

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 The closeness of a measured value to the true value is called:

- A) Precision B) Sensitivity
☒ C) Accuracy D) Reproducibility

PMMC instruments can be used for:

- A) AC only ☒ B) DC only
C) AC and DC D) High frequency only

Moving Iron instruments operate on:

- A) Electrostatic effect B) Electromagnetic induction
☒ C) Magnetic effect of current D) Hall effect

4 The driving torque in an induction energy meter is produced by:

- A) Eddy currents ☒ B) Permanent magnet
C) Spring action D) Electrostatic force

5 The braking torque in an induction energy meter is produced by:

- A) Spring B) Air friction
☒ C) Permanent magnet D) Electromagnet

Kelvin double bridge is used for measuring:

- A) High resistance ☒ B) Medium resistance
☒ C) Low resistance D) Capacitance

Which AC bridge is used for measurement of capacitance and dielectric loss?

- A) Anderson Bridge ☒ B) Schering Bridge
C) Maxwell Bridge D) Wien Bridge

8 A Current Transformer should never be operated with:

- A) Burden connected B) Open secondary
C) Short-circuited secondary ☒ D) Ammeter connected

9 The Hall Effect sensor is used for measuring:

- A) Magnetic field ☒ B) Humidity
C) Light intensity D) Sound

- 6
- 10 The output of a thermocouple is based on:
 A) Hall effect B) Seebeck effect
 C) Piezoelectric effect D) Faraday's law
- 11 The instrument used to measure quality factor (Q) of coils is:
 A) CRO B) LCR Meter C) Q-Meter D) DVM
- 12 Automatic Meter Reading (AMR) is mainly used for:
 A) Manual billing B) Remote meter data collection
 C) Power generation D) Voltage regulation
- Q2. Solve the following.
- A) Define accuracy, precision, sensitivity, tolerance, and reproducibility. Explain their significance in measurement systems. 6
- B) Explain the construction and working principle of a PMMC instrument. Derive the expression for deflecting torque. 6
- Q3. Solve the following.
- A) Explain the methods of measuring active and reactive power in a balanced three-phase circuit. 6
- B) Discuss Kelvin's double bridge for measurement of low resistance and derive the balance equation. 6
- Q4. Solve Any Two of the following.
- A) Explain the construction and working of a single-phase induction type energy meter. Discuss creeping and braking adjustments. 6
- B) Explain the construction and operation of Schering bridge. State its applications and limitations. 6
- C) A Maxwell bridge has standard capacitor $C = 0.5 \mu\text{F}$, resistance $R_2 = 500 \Omega$, and $R_3 = 1000 \Omega$. Determine the unknown inductance at balance condition. 6
- Q5. Solve Any Two of the following.
- A) Explain the construction and working of a Current Transformer (CT). Define ratio error and phase angle error. 6
- B) Describe the working principle, characteristics, and industrial applications of Hall Effect sensors and piezoelectric transducers. <https://www.batuonline.com> 6
- C) Explain the construction, working principle, and applications of LVDT and strain gauge transducers. 6
- Q6. Solve Any Two of the following.
- A) Explain the block diagram and working principle of a Digital Voltmeter (DVM). 6
- B) Explain the construction and operation of a Digital Storage Oscilloscope (DSO). State its advantages over CRO. 6
- C) Explain the block diagram, working principle, and Automatic Meter Reading (AMR) technique used in Smart Energy Meters. 6

*** End ***